



CS 540 Module Three Activity Guidelines and Rubric

Overview

Data scientists and analysts frequently work with data sets from multiple systems, departments, or tools. These data sets are rarely analysis-ready and often include inconsistencies, missing values, or formatting issues. Cleaning and merging data is a critical step in building reliable, structured data models that support meaningful insights. This work mirrors what professionals do when preparing data for analysis, machine learning, or reporting. Mastering this process improves your ability to manage real-world data challenges and ensures your models are built on accurate and well-organized information.

Scenario

You are a junior data analyst on a data team analyzing weather data. Each platform exports user data in a slightly different format, with inconsistencies in naming, missing entries, and duplicate records. In addition to preparing this data for a retention and segmentation strategy, your team has been asked to evaluate potential risk mitigation strategies based on environmental factors that could impact the company's data infrastructure.

Specifically, leadership wants to better understand how weather events such as hurricanes, floods, or extreme temperatures could affect the placement and operation of data centers. You will incorporate location-based weather pattern data alongside the engagement data to support recommendations for infrastructure resilience.

As the junior data analyst on the team, you must use Python to clean and blend multiple data sources into one structured data set.

Directions

In this activity, you will write Python code to clean and merge data sets, forming a unified and structured data model. You are expected to apply standard preprocessing techniques and write clear, functional code that demonstrates your understanding of each step. When you are ready to begin your work, access Codio by clicking the title of the next task. Choose two to three data sets located in the Module Three Activity data folder in Codio for your assignment.

When the assignment is complete, download the HTML version of your project by clicking on **File**, then **Download As**, then **HTML (.html)** from the Jupyter menu bar. Save the file to your local machine.

Note: if the code block was not run in the Notebook, it may not render in the HTML output of the Notebook. To fix missing blocks, run the code block, then download as HTML.

Specifically, you must address the following rubric criteria:

Part One: Python Code

1. **Inspect** each data set to identify issues. Consider the following:

- A. Are there any missing values?
- B. Are there issues with inconsistent formatting?

C. Are there irrelevant columns?

2. Clean each data set using appropriate **preprocessing techniques**. Consider the following:

- A. Renaming columns
- B. Dropping nulls or duplicates
- C. Converting data types
- D. Normalizing values

3. Explain your **rationale** for each technique or transformation. Include the following:

- A. Inline comments of your cleaning decisions

Part Two: Merged Data

4. Merge the data sets into a **cohesive structure** using appropriate join methods. Consider the following:

- A. Have you ensured alignment across shared keys and maintained consistent formatting?

5. **Validate** your results. Consider the following:

- A. The number of rows, columns, and any remaining inconsistencies

What to Submit

To complete this activity, you must submit the following:

Python Code

Download your Jupyter Notebook as an HTML file. Review the file to ensure the code and output are displayed. Submit the HTML file.

Merged Data

Download your csv output file of your merged data. Review the file to validate your results in the merged data output. Submit the csv file.

AI Usage

If you use gen AI tools to support your work on this assignment, be sure to follow [these AI usage guidelines](#). You must acknowledge your use of these tools in your work. Guidelines on how to cite AI tools can be found in [this Shapiro Library guide](#).

Module Three Activity Rubric

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Part One: Inspect	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Inspects each data set to identify issues	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Part One: Preprocessing Techniques	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Cleans each data set using appropriate preprocessing techniques	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Part One: Rationale	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Explains the rationale for each technique or transformation, including inline comments of the cleaning decisions	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Part Two: Cohesive Structure	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Merges the data sets into a cohesive structure using appropriate join methods	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Part Two: Validate	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Validates the results	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Clear Communication	Exceeds expectations with an intentional use of language that promotes a thorough understanding	Consistently and effectively communicates in an organized way to a specific audience	Shows progress toward meeting expectations, but communication is inconsistent or ineffective in a way that negatively impacts understanding	Shows no evidence of consistent, effective, or organized communication	5
Total:					100%